

Introduction

❑ Overview:

- This Mail- is an advanced email client designed to declutter communication through Artificial Intelligence and Automation along with analytics for efficiency.

❑ Purpose:

- To bridge the gap between traditional email protocols (IMAP/SMTP) and modern productivity needs.
- Users often face inbox overload, missed deadlines, and security threats.
- The Email Manager System aims to intelligently organize and manage emails.

❑ Core Focus:

- Improving efficiency by combining multi-account management, AI-driven filtering, and security auditing into one platform.

❑ Target Audience:

- Professionals, Students, Business teams, etc.

Problem Statement

In today's digital environment, email remains one of the most essential communication tools for students, professionals, and organizations. However, with the increasing volume of daily emails, users often struggle to stay organized, respond on time, and identify what truly matters.

❑ Problem Statement

Most users face challenges such as:

- **Inbox Overload:** Managing large volumes of emails becomes time-consuming and overwhelming.
- **Missing Important Emails:** High-priority emails often get buried under less relevant messages.
- **Poor Organization:** Attachments, deadlines, and follow-ups are difficult to track manually.
- **Lack of Automation:** Existing clients provide basic features but lack personalized prioritization, reminders, and analytics.
- **Security Concerns :** Detecting suspicious or phishing emails manually is unreliable and risky.
- There is a clear need for a smart, unified system that can **prioritize, organize, secure, and analyze mails automatically**, helping users stay efficient and focused.

Project objective

❑ Primary Goal:

- To develop an intelligent Email Optimization System that automates email prioritization, organization, and security, helping users manage their inbox efficiently with minimal manual effort.

❑ Specific Objectives:

- Automatically classify incoming emails into priority categories.
- Detect and highlight urgent or deadline-based emails.
- Provide reminders and follow-up alerts for un-replied messages.
- Organize email attachments by type, sender, and date.
- Offer a clean, unified dashboard for viewing and managing emails.
- Provide basic phishing/suspicious email detection to enhance user safety.
- Enable analytics and insights such as email frequency, usage patterns, and sender trends.

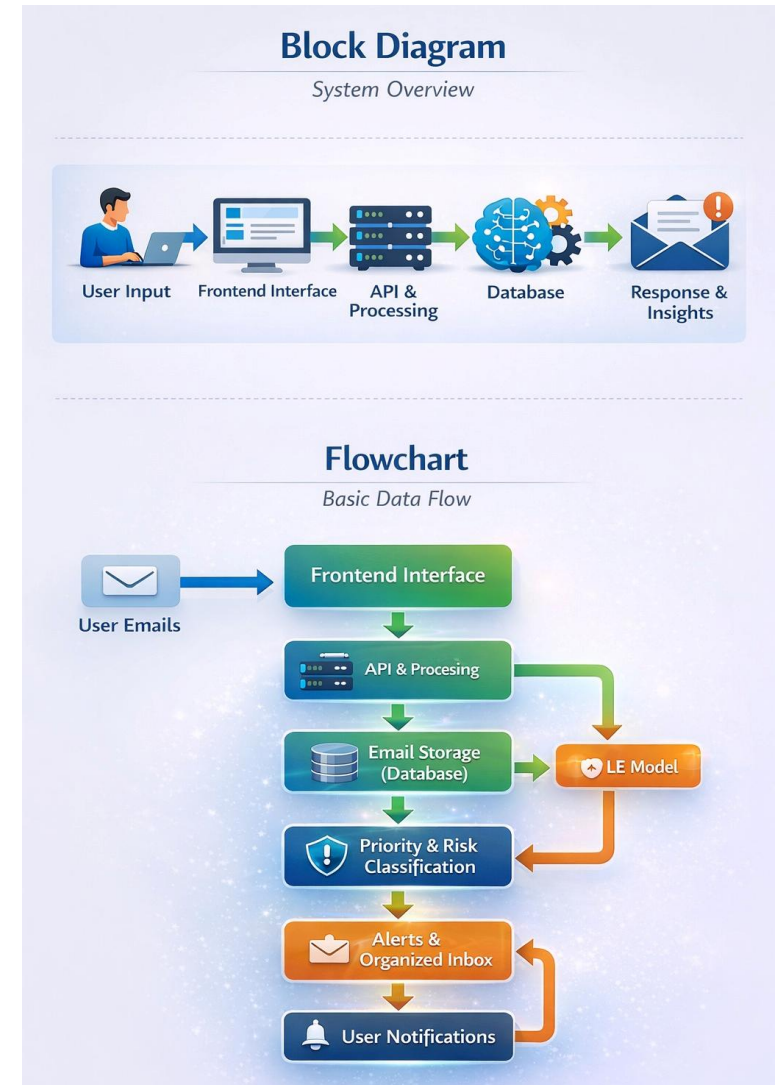
❑ Scope Boundary:

- The system is limited to single-user, read-only email management using basic ML/rule-based methods, with no email sending, advanced security, deep learning, cross-platform integration, cloud scalability, or legal compliance features.

Proposed Solution

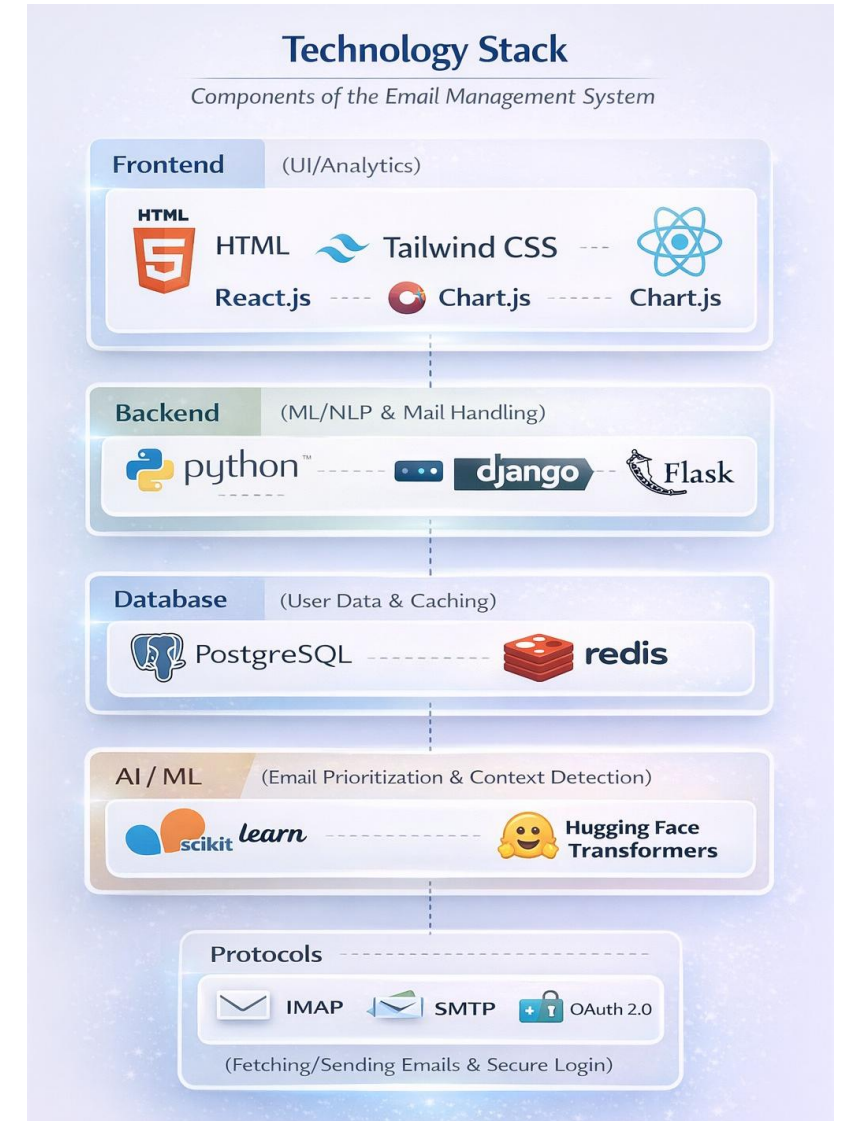
□ Overview of the Solution:

- Our Smart Mail Optimizer System provides an intelligent and unified platform that prioritizes, organizes, and analyzes emails automatically.
- The system integrates a smart backend using rule-based logic and lightweight ML models to improve email management efficiency, reduce inbox clutter, and enhance user productivity.
- An intelligent Email Manager with a clean web-based dashboard.
- Backend processes emails using rules and basic ML models.
- ML/NLP modules classify priority, urgency, and security risk.
- Users receive alerts, insights, and organized email views.



Technology Stack

- ❑ **Frontend** : HTML, Tailwind CSS (styling), React.js (for a responsive, single-page UI), Chart.js (analytics)
- ❑ **Backend** : Python (Django or Flask) – chosen for its extensive libraries in NLP and Mail handling
- ❑ **Database** : PostgreSQL (user data & logs), Redis (caching for speed).
- ❑ **AI/ML** : Scikit-learn (Naive Bayes for prioritization), HuggingFace Transformers (Context detection).
- ❑ **Protocols** : IMAP (fetching), SMTP (sending), OAuth 2.0 (secure login).



Feasibility Study

❑ Technical Feasibility

- Our team is proficient in **Python, Node.js, web dev**, and basic **machine learning**, which are sufficient to build the Email Manager System.
- Features like email fetching (IMAP/SMTP), rule-based filtering, and lightweight ML models fall well within our technical capabilities.
- No requirement for advanced AI, DL, or complex system architecture, making the project technically achievable within our skill set.

❑ Resource Feasibility:

- **Software & APIs:** All required tools (Python, Node.js, MongoDB, Scikit-learn) are **open-source and free**.
- Email access via **IMAP/SMTP** requires no paid APIs.

❑ Data Availability

- **Email data** is fetched directly from the user's inbox via IMAP, so no external dataset is needed.
- For training simple ML models (spam detection or priority classification), datasets can be sourced from:
 - **Kaggle public email/spam datasets**
 - **Enron Email Dataset (open-source)**
- No data licensing issues since we only use publicly available datasets.

Tentative Timeline (Gantt Chart)

□ Project Phase and Timeline:

Phase	Tasks	Week
1. Requirement Analysis	Finalize features, system scope, tech stack	Week 1
2. System Design	Architecture design, UI wireframes, database schema	Week 2
3. Frontend Development	Dashboard layout, inbox UI, reminder/analytics pages	Week 3 – Week 4
4. Backend Development	IMAP/SMTP integration, APIs, database setup	Week 4 – Week 5
5. ML/Rule-Based Engine	Priority classification, spam/phishing detection	Week 5 – Week 6
6. Integration Phase	Connect frontend ↔ backend ↔ ML components	Week 6
7. Testing & Debugging	Unit tests, functionality tests, user testing	Week 7
8. Final Deployment	Deployment on server, final review, presentation prep	Week 8

Expected Outcomes

❑ Final Deliverables

- **A fully functional web-based Email Manager System** featuring:
 - Smart email prioritization
 - Reminder and follow-up alerts
 - Attachment organizer
 - Basic phishing/suspicious email detection
 - Email analytics dashboard
- **Backend APIs** for email fetching, processing, and analytics
- **Lightweight ML models** for classification and spam detection
- **Project documentation**, including system design, architecture, and testing reports

❑ Beneficiaries

- **Students** who struggle with managing academic or internship-related emails
- **Working professionals** handling high volumes of daily emails
- **Small teams / startups** needing simple email organization tools
- **Anyone** wanting a smart, efficient, distraction-free email management experience